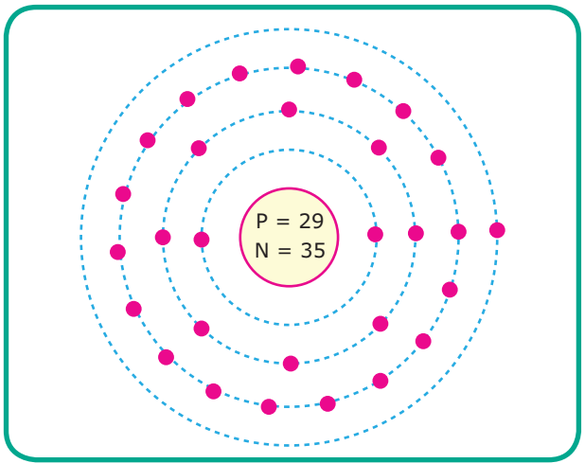


Figure 3.2 Atomic structure of copper

The Figure 3.2 shows the structure of copper atom. It has 29 electrons which are arranged in different orbits as follows. The first orbit will have 2 electrons, the second 8 electrons, the third 18 electrons and fourth orbit will have 1 electron. The atomic structure of all known elements can be shown in this way and we can refer few elements.

3.2.1 Electron

Since electronics deals with tiny particles called electrons, these small particles require detailed study. *An electron is a negatively charged particle having negligible mass.* Some of the important properties of an electron are:

1. Charge of an electron,
   $e = 1.602 \times 10^{-19}$ coulomb
2. Mass of an electron,
   $n = 9.0 \times 10^{-31}$ kg
3. Radius of an electron,
   $r = 1.9 \times 10^{-15}$ meter

3.2.2 Energy of an Electron

An electron moving around nucleus possesses two types of energies, viz.

Kinetic Energy -due to its motion (relativity)
Potential Energy - due to the charge in the nucleus

The total energy of the electron is the sum of these two energies. The total energy of the electron increases as its distance from the nucleus increases. Hence, the electron in the last orbit possesses high energy than the electrons in the previous orbits. The last orbit electron plays important role in determining the physical, chemical and electrical properties of a material.

Valence Electron

The electrons in the outermost orbit of an atom are known as valence electrons.

The outermost orbit can have maximum of 8 electrons. i.e., the maximum number of valence electrons can be 8. The valence electron determines the physical, chemical and electrical properties of material.

3.2.3 Atomic Structure of Materials

On the basis of electrical conductivity, materials are generally classified into conductors, insulators and semi-conductors. In general one can determine the electrical behaviour of a material from the number of valence electron as under.

i. Conductor:

When the number of valence electron of an atom is less than 4 (i.e., half of the maximum of 8 electron) the material is usually a metal and a conductor. Examples are sodium, magnesium and aluminium which have 1,2 and 3 valence electrons respectively. Is shown in Figure 3.3.

ii. Insulator:

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